

Revision 0: Requirements and Hazard Analysis

Group 8 - C.A.V.E: Cave Assessment and Visualization Equipment

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Table 0: Revision History

Date	Developer(s)	Change
November 16, 2025	All Members	Initial Draft

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1 Introduction

1.1 Purpose

The purpose of this document is to outline the system requirements of the C.A.V.E. system which will highlight expected behaviours like what inputs are collected, what type of processing is done on the data, and what output is being created. It will define all the necessary variables and constants used to achieve this, as well as any notation or technical terms that are needed. There will also be a fault tree analysis (FTA) breakdown of the possible root causes of failure that will inform the significant safety requirements of the system.

1.2 Scope

The scope of the system includes the development of a portable or wearable 3D data-acquisition device capable of reconstructing a scanned environment with integrated sensors and processing algorithms. The system records sequential point clouds from the Time-of-Flight sensors, fuses IMU orientation data and processes the collected data to generate an accurate and reliable 3D scan of the environment. This scope covers the areas of data capture, pre-processing, point cloud registration, as well hardware and software integration.

2 Project Definitions & Variables

2.1 Glossary

The following table consists of definitions of certain terms and notations that are referenced in the project.

Term	Definition
IMU	Inertial Measurement Unit → Provides acceleration and rotational data
LiDAR	Light Detection and Ranging → A sensor that uses a rotating laser to measure distances to objects
Point Cloud	A collection of coordinate points in 3D space
Portable Device	The portion of the design that is carried with the user/operator for mapping an enclosed space
TOF	Time-of-Flight depth sensor → Uses pulses of infrared light and a sensor that measures the time at which the light is received back to create a point cloud

Table 1: Definitions

2.2 Monitored and Controlled Variables

Monitored variables are information from the environment being input to the system, and controlled variables are the output products of the system's internal processes. The following tables list and describe these variables in the C.A.V.E. system.

Variable	Description	Unit	Purpose
M_tof_data	A single depth image frame captured from the environment	Array of depths and positions in metres	Capturing the contours and shape of the environment
M_imu_gyro	Device rotation rate about x,y,z axes, measured by IMU	Degrees/second tuple ($\text{deg}/s_x, \text{deg}/s_y, \text{deg}/s_z$)	Provide reference for position change between M_tof_data captures
M_imu_accel	Device acceleration in x, y, z axes, measured by IMU	Metres/second ² tuple ($m/s_x^2, m/s_y^2, m/s_z^2$)	Additional frame translation reference
M_imu_magnet	Change in magnetic field	Tesla tuple (T_x, T_y, T_z)	Additional frame rotation reference
M_lidar	2D Li-DAR scan of the environment	Tuple of distance in metres and angle in degrees	Long range slice of the environment to aid convergence
M_start_stop	User toggle enabling or disabling the system	Boolean, true == on	Allow user control of starting and stopping a mapping session
M_SOC	Remaining battery percentage	Percentage of total charge (%)	Estimate remaining time available for mapping
M_storage_left	Remaining storage space available for data capture	Percentage of allocated storage (%)	Estimate remaining storage available for mapping
C_record_status	User facing indicator to show the status of M_start_stop	Boolean	Inform user of the status of M_start_stop
C_record_remain	User facing indicator to show how long recording can continue for, given M_SOC and M_storage	Integer in the form MM:SS	Inform user how much longer they can record
C_point_cloud	Final point cloud or 3D representation of the merging of all captured data	Point cloud; array of X,Y,Z tuples	Main product of the device, 3-dimensional representation of the mapped area (cave)

C_confidence	Indicator of the quality of the C_point_cloud representing the proportion of successfully merged capture instants	Percentage (%)	Show how well C_point_cloud represents the captured area
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Table 2: Variables and its description

2.3 Constants

This section describes the values that will not change during the operation of the device and should be set to a specific value during design.

Variable	Description	Unit	Purpose
Const_transform_i	Transformation matrix of sensor _i from the reference frame (likely chosen as the IMU or centre of portable device)	4x4 matrix representing translation and rotation in 3D	Define spatial relationship between different sensor's captures
Const_poll_rate_i	Rate at which sensor _i makes information available to the system	Hertz	Inform rate of data ingest
Const_capt_size_i	Size of a single capture of information by sensor _i	Gigabytes (GB)	Inform Const_datarate
Const_capt_rate	Reciprocal of the interval between a total system capture of all sensors	Hertz (s ⁻¹)	Inform Const_datarate
Const_storage	Storage capacity of the portable device	Gigabytes (GB)	Inform M_storage_left
Cosnt_datarate	Total ingested data from all sensors, the sum of Const_capt_rate * Const_capt_size_i	Gigabytes/second (GB/s)	Inform C_record_remain
Cosnt_power_rate	Average power consumption rate of the portable device	Watts (W)	Informs C_record_remain
Cosnt_power_cap	Battery capacity of the portable device	Watt-hours (Wh)	Informs C_record_remain

Table 3: Constants and its description

3 System Diagram

In Figure 1, the system is represented as a black-box, where the monitored variables serve as the inputs to the system and the controlled variables are the resulting outputs.

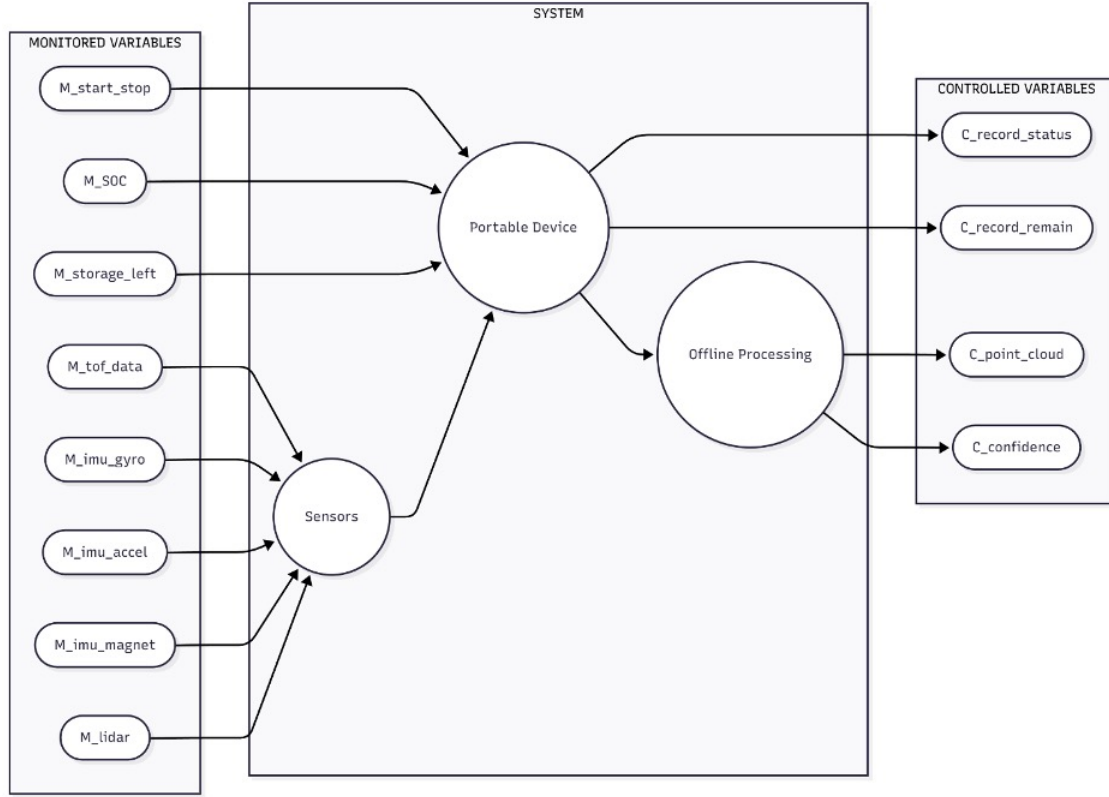


Figure 1: Representation of system boundaries

4 Functional Decomposition

The following figure shows the functional decomposition of the C.A.V.E. system.

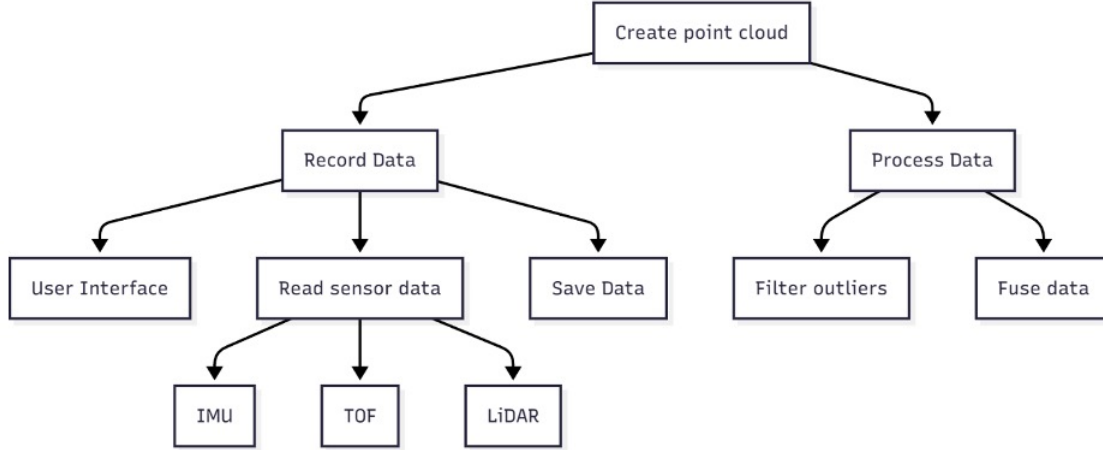


Figure 2: Functional decomposition of the system

5 System Requirements

Requirement Type	Likelihood of Change
Behavioural	
RB1: The device will produce a point cloud <code>C_point_cloud</code> representing the physical surfaces mapped by the sensors (<code>M_tof_data</code> , <code>M_lidar_data</code>) while recording is enabled (<code>M_start_stop == true</code>).	Low - this is a core functionality of the device.
RB2: The confidence in the accuracy of <code>C_point_cloud</code> shall be represented by <code>C_confidence</code>	Possible - It may be hard to produce a useful measure of <code>C_confidence</code> .
RB3: <code>C_record_status</code> toggles between indicating recording and standby when <code>M_start_stop</code> is toggled by the user. If the system encounters an error it cannot recover from, it must display an error status on <code>C_record_status</code> .	Low - this is a basic error communication.
RB4: <code>C_record_time</code> will display remaining available recording time by taking the minimum of <code>M_SOC/const_power_rate</code> and <code>M_storage_left/const_datarate</code> .	Low - Important to achieve for ease of use by the operator.
RB5: The system shall maintain a <code>C_confidence</code> $> 50\%$ in little to no visible light conditions.	Possible - This is a core function but the exact confidence measurement could change.
Safety	
RS1: The battery shall be protected from punctures, overheating, or shorts.	Low - this is a basic safety measure.
RS2: The device shall not expose any dangerous voltages to the user.	Low - Also a basic safety measure.

Timing	
RT1: Const_capt_rate shall be frequent enough to produce complete maps of an environment at a walking pace (3-5km/h)	Low - Core function needed for ease of use.
RT2: Const_capt_rate shall be low enough such that the system can complete a measurement of all sensor data before the next interval.	Low - Need to ensure consistent data.
Hardware/Software Environment	
RE1: The portable device shall not require an active internet connection to achieve the behavioural requirements RBX.	Low - Core function, must potentially operate underground.
RE2: The hardware should minimize encumbrance of the user to ensure exploration is not inhibited.	Low - Important for ease of use.

Table 4: Requirement and likeliness to change

6 Hazard Analysis

In this section, the safety requirements are determined by breakdown of the possible hazards that may occur in our system with respect to both software and hardware components. This is presented in an FTA in Figures 3 and 4.

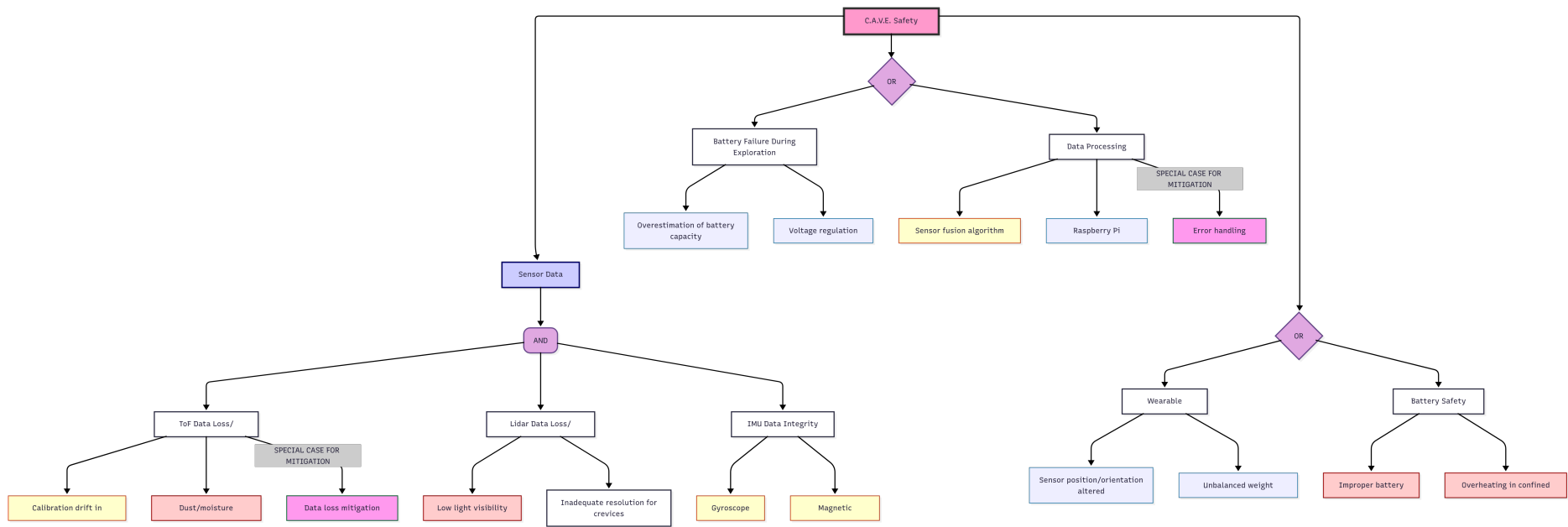


Figure 3: FTA of the software components in the system

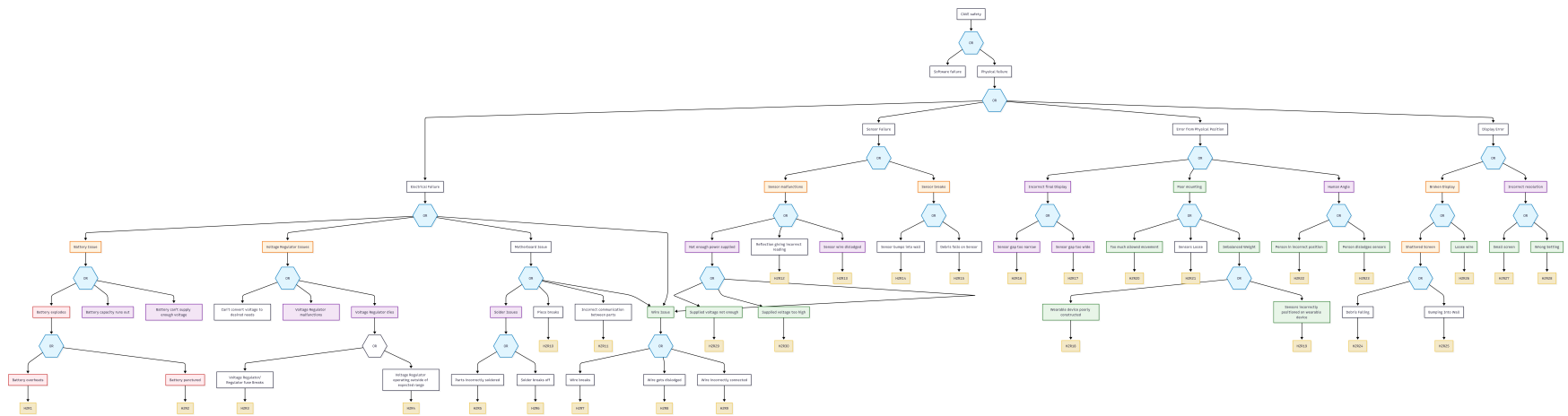


Figure 4: FTA of the hardware components in the system

6.1 Error Handling/Mitigation

The following table lists the error handling method or mitigation when any hazards occur in the system along with it respective severity.

Hazard ID	Description	Severity	Response Strategy	Mitigation Plan
H2R1	The battery overheats	High	Mitigate	Proper cooling, proper rated battery
H2R2	Battery punctured	High	Avoid	Proper casing, kept in a place unlikely to get impacted
H2R3	Voltage regulator fuse breaks	Medium	Mitigate	Proper rating of a regulator so it doesn't fall outside of expected range
H2R4	Voltage regulator operating outside of expected range	Medium	Mitigate	Make sure no sudden power draws occur
H2R5	Parts incorrectly soldered	Medium	Avoid	Double-check soldered equipment
H2R6	Solder breaks off	Medium	Avoid	Use ample amount of solder
H2R7	Wire breaks	Medium	Avoid	Ensure wire is taut and flexible, double-check connections
H2R8	Wire gets dislodged	Medium	Avoid	Double-check wire is securely connected to different pieces, stress-test connection
H2R9	Wire improperly connected	Medium	Avoid	Double-check what the wires are connecting, make sure it follows a plan
H2R10	Motherboard component breaks	Medium	Avoid	Keep motherboard secure and in a place less prone to bumps/potentially breaking
H2R11	Motherboard parts incorrectly communicate	Medium	Avoid	Double-check motherboard connections, ensure everything is wired as expected.
H2R12	TOF sensor providing point cloud of a reflection (e.g water)	Medium	Mitigate	Ensure the other sensor can catch errors and correct them in post

H2R13	Sensor wire dislodged	Medium	Avoid	Provide the sensor enough support so the wire wouldn't accidentally break or get damaged
H2R14	Sensor bumps into wall	Medium	Avoid	Ensure sensor has proper protection so it's not directly in contact with cave walls
H2R15	Debris falls on sensor	Medium	Avoid	Ensure sensor is inset, or protected from potential vertical hazards
H2R16	Sensor gap too narrow	Medium	Avoid	Correct planning to ensure optimal sensor placements such that there is minimal overlap to produce efficient imaging
H2R17	Sensor gap too wide	Medium	Avoid	Correct planning to ensure optimal sensor placements such that there is minimal overlap to produce efficient imaging
H2R18	Wearable device poorly constructed	Low	Avoid	Proper simulation before printing, as well as rigorous testing to make sure it's constructed well
H2R19	Sensors incorrectly positioned	Low	Avoid	Mirror sensor positions so that weight is balanced
H2R20	Too much allowed movement (for sensors)	Low	Avoid	Limit degrees of freedom to 0 or 1 for each sensor
H2R21	Sensors Loose	Medium	Avoid	Proper machining such that the sensors fit snugly, without enough space to wiggle around.
H2R22	Person in incorrect position	Low	Accept	Explain to the user how to be properly angled, but accept there will be some human error in the beginning
H2R23	Person dislodges sensor	Low	Accept	Understand that there are human errors, and they may push some things incorrectly. Proper machining and construction will mitigate this as much as possible

H2R24	Debris Falling, shattering screen	Medium	Mitigate	Give the screen a protector so that it wouldn't be harmed to a degree of breaking
H2R25	Bumping into a wall, shattering screen	Medium	Mitigate	Give the screen proper protection so it wouldn't get damaged to such a degree
H2R26	Loose wire for the display	Low	Avoid	Proper connection between the display and other hardware, make sure there is enough wire such that it doesn't snap or pull off
H2R27	Screen too small	Low	Avoid	Research prior for assumed size of screen, such that everything would fit as expected
H2R28	Screen resolution incorrect	Low	Avoid	Make sure the screen used supports the resolution we have to display properly
H2R29	Under-supplied voltage	Low	Avoid	Shut down system if not enough power
H2R30	Over-supplied voltage	Low	Avoid	Shut down system if prolonged over-supplied voltage, to make sure parts don't break

Table 5: Hazards and its mitigation plan